

Math 54 Discussion Worksheet

Name: _____

ID: _____

Problem 1

- a) Show that the matrix

$$A = \begin{bmatrix} 2 & 1 & 6 \\ 0 & 2 & 5 \\ 0 & 0 & 2 \end{bmatrix}$$

has $r = 2$ as a repeated eigenvalue with multiplicity 3. Also, show that the eigenvectors of A are all of the form $u = \text{scol}(1, 0, 0)$.

- b) Using the result above, find a solution to the system $x' = Ax$ of the form $x_1(t) = e^{2t}u_1$.
- c) Get a second linearly independent solution by considering $x_2(t) = te^{2t}u_1 + e^{2t}u_2$.
- d) Get a third linearly independent solution by trying $x_3(t) = \frac{t^2}{2}e^{2t}u_1 + te^{2t}u_2 + e^{2t}u_3$.
- e) Show that $(A - 2I)^2u_2 = (A - 2I)^3u_3 = 0$.

Problem 2

Using matrix algebra techniques, find a general solution to the system

$$x' = 3x - 4y$$

$$y' = 4x - 7y$$

Problem 3

Show that $x_1(t)$ and $x_2(t)$ given by

$$x_1(t) = e^{\alpha t} \cos(\beta t) \mathbf{a} - e^{\alpha t} \sin(\beta t) \mathbf{b}$$

$$x_2(t) = e^{\alpha t} \sin(\beta t) \mathbf{a} + e^{\alpha t} \cos(\beta t) \mathbf{b}$$

are linearly independent provided that $\beta \neq 0$ and that $\mathbf{a}$ and $\mathbf{b}$ are not the zero vector.